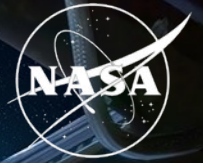


National Aeronautics and
Space Administration



Interpretable and generative models for mouse spaceflight

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Biological & Physical Sciences



PROJECT

One dataset, two machine-learning questions

We used mouse bulk RNA-seq to study how spaceflight changes genes, pathways, and tissue state.

NASA OSDR API Mus musculus bulk RNA-seq | spaceflight (FLT) and ground control (GC) | multiple missions and tissues

01 Which biological programs change?

expiMap

Train a pathway-constrained reference on ARCHS4, then map OSDR samples into mouse Reactome programs.

Output: tissue-specific pathway shifts

02 Can synthetic profiles sharpen the analysis?

Conditional generation

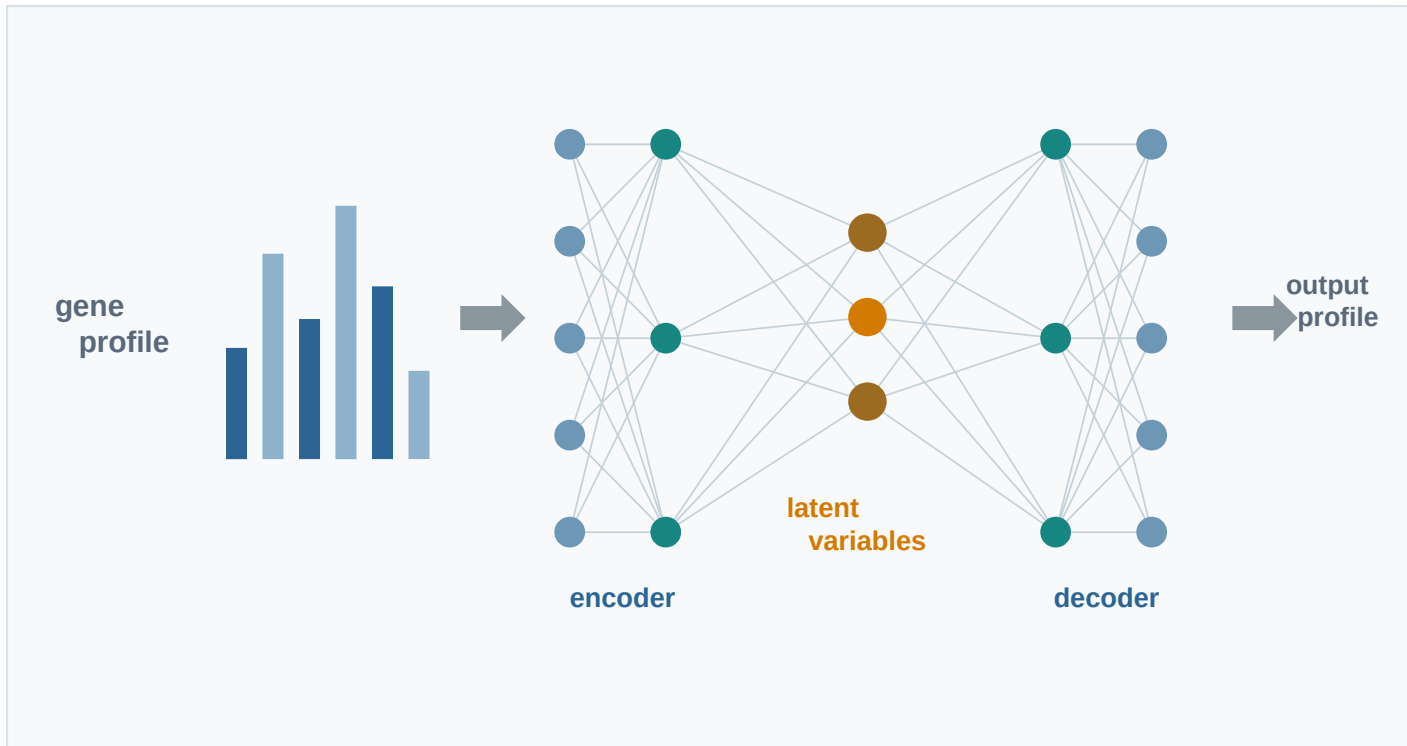
Generate matched expression profiles by tissue, study context, and FLT or GC condition.

Output: validated profiles and gene ranking

Autoencoders compress expression into a latent space

The encoder summarizes a gene profile; the decoder learns enough structure to reconstruct it.

AUTOENCODER



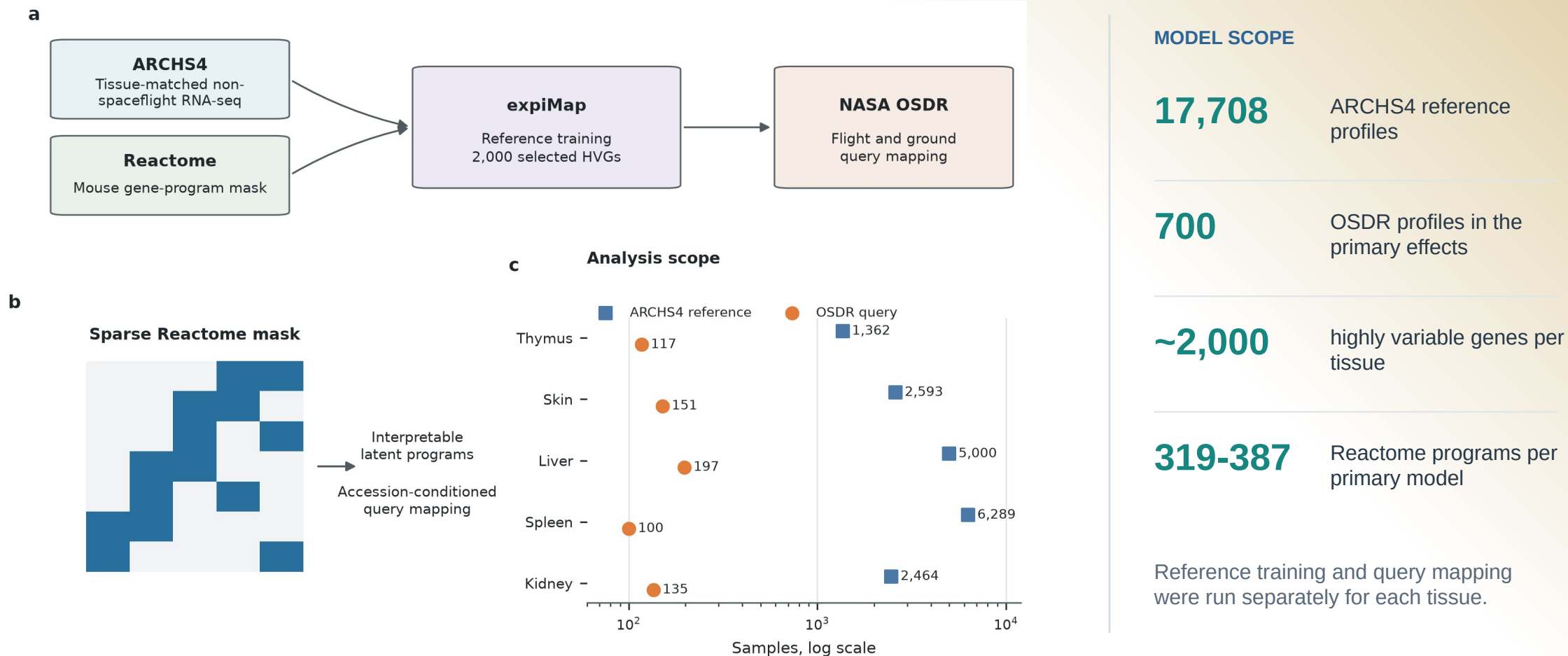
LATENT SPACE



expiMap: each constrained latent variable corresponds to a Reactome gene program.

Reactome pathways make the latent space interpretable

Tissue-matched ARCHS4 references define the background; OSDR FLT and GC samples are mapped as queries.



EXPIMAP RESULTS

Four tissues produced the clearest pathway patterns

Effects were estimated within each project before the project-level shifts were combined.

Thymus

117 samples | 5 projects

LOWER IN FLIGHT

DNA repair; RHOA cytoskeletal cycle; lymphoid-stromal interactions

Known involution may also involve lower repair, motility, and niche coordination.

Skin

151 samples | 4 projects

LOWER IN FLIGHT

Chromatin regulation; DNA repair; Hedgehog; sphingolipids; cell junctions

Barrier injury may include a broader loss of tissue maintenance and coordination.

Liver

197 samples | 9 projects

LOWER IN FLIGHT

MHC class II antigen presentation; T-cell receptor signaling

Metabolic heterogeneity coexisted with lower adaptive immune communication.

Spleen

100 samples | 5 projects

LOWER IN FLIGHT

T-cell receptor signaling; neutrophil degranulation; C-type lectin signaling

The most consistent multi-pathway result joined adaptive and innate immune changes.

BACKGROUND

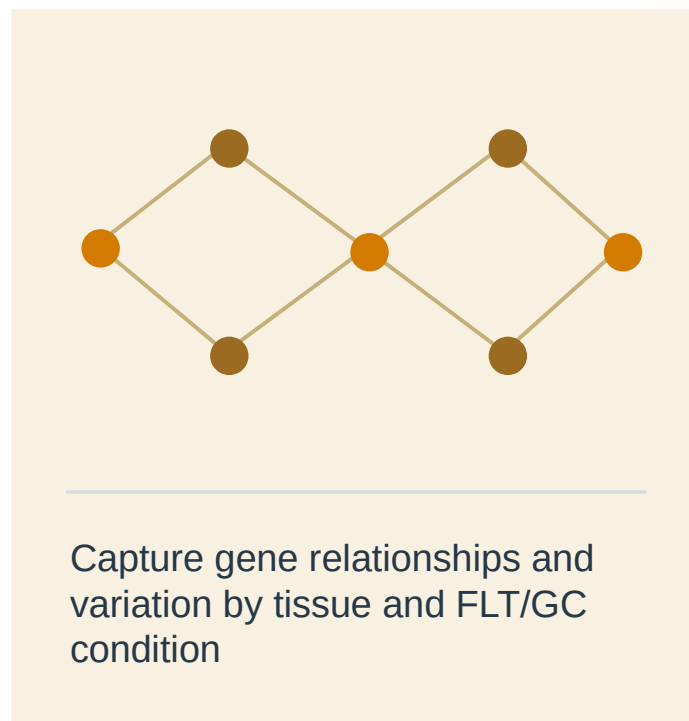
What is synthetic transcriptomics?

A generator learns patterns in measured RNA-seq data, then samples new expression vectors under chosen conditions.

01 Observed profiles



02 Learn the distribution



03 Synthetic profiles



QUESTION

Small studies and study effects complicate tissue comparisons

Can a generator help rank spaceflight signal without pretending it creates new animals?

NASA OSDR

Observed spaceflight cohort

1,610

profiles

75

accessions

835 FLT

775 GC



ARCHS4 mouse

Reference pretraining cohort

997,515

profiles screened



17,244

selected

20 tissues

Study effects can look like spaceflight biology.

- 01 Preserve tissue and FLT/GC structure.
- 02 Avoid memorizing individual profiles.
- 03 Test biological effects in observed OSDR samples.

METHOD

We built a configurable bulk RNA-seq generation pipeline

The downstream analysis used the outlined options; the remaining choices stay configurable.

01 Data scope

Sources

OSDR API

ARCHS4

Studies

Single

Multiple

Tissues

Per tissue

All tissues

02 Transform

Expression

Raw

CPM

TPM

Scaling

None

Z-score

MaxAbs

Features

All

HVG

L1000

03 Harmonization

Global or study correction

None

Within-study z-score

ComBat / MBatch

MOBER

04 Model training

Generator

WGAN-GP

DDIM

Training source

OSDR only

ARCHS4 only

Pretrain + adapt

05 Conditioning

Model inputs

Tissue

FLT / GC

Accession

Material type

Sex / age

Downstream

configuration

PREPROCESS

TPM / MaxAbs / 974 landmarks

PRETRAIN

ARCHS4

ADAPT

OSDR

GENERATE

Conditional DDIM

CONDITION

Tissue

FLT / GC

Accession

Material type

HARMONIZE

None

SCOPE

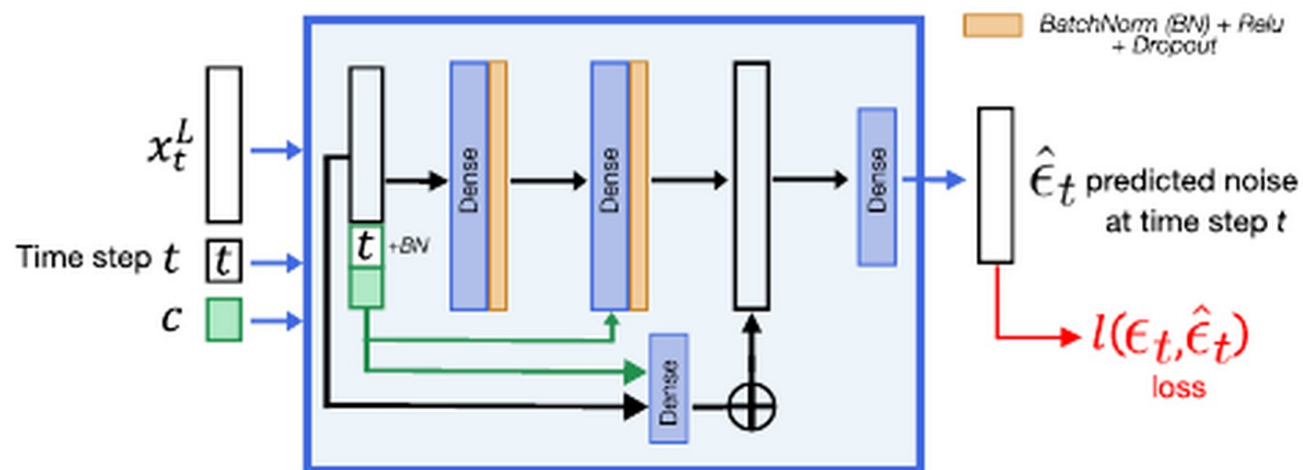
All tissues

VALIDATION

DDIM had lower separability and distributional distance

Both generators matched expression. DDIM was harder to distinguish from real profiles and had higher F1.

Architecture from Lacan et al.



ARCHS4 tissue probe

Real 0.781 | Synthetic 0.781

Balanced accuracy is preserved in generated tissue labels.

Metric	WGAN-GP	DDIM	Read
Correlation	0.976	0.974	similar
F1	0.985	0.997	higher
Adversarial acc.	0.636	0.475	near 0.5
FD / real P95	0.144	0.074	lower

Use DDIM

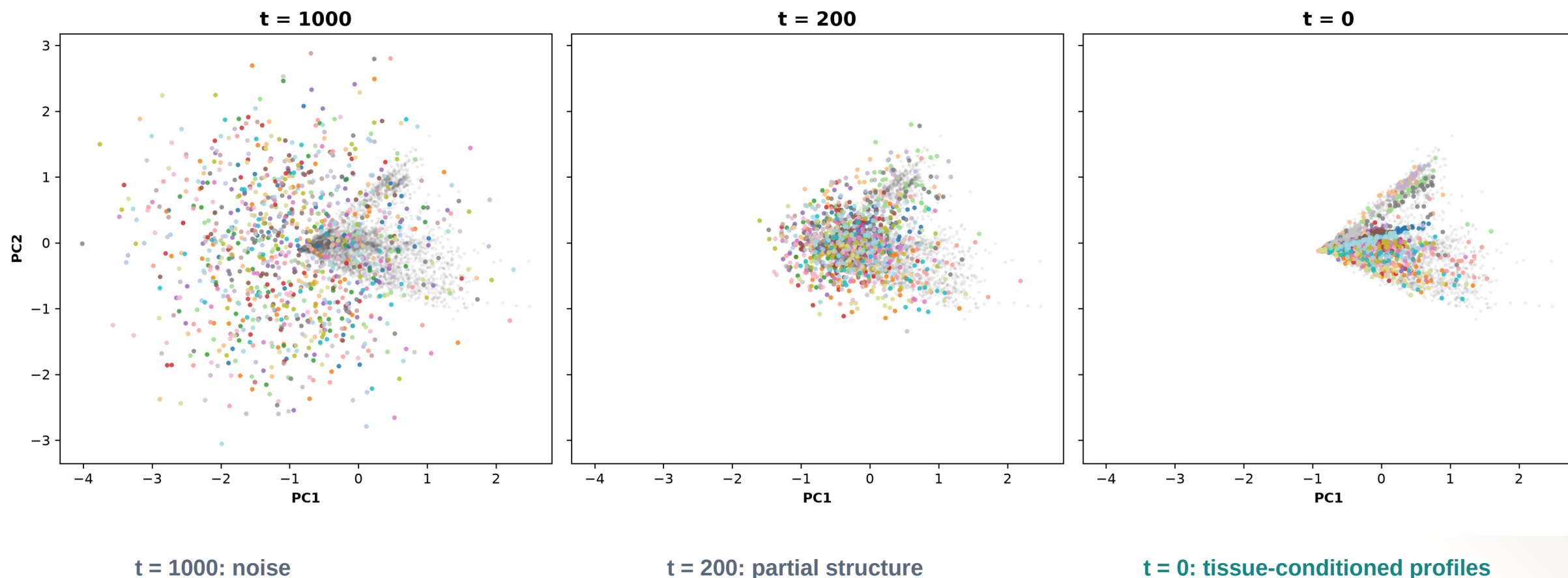
Higher F1 with lower separability and FD.

AA near 0.5 means real and generated profiles are difficult to separate.

DIFFUSION

Diffusion learns tissue structure from noise

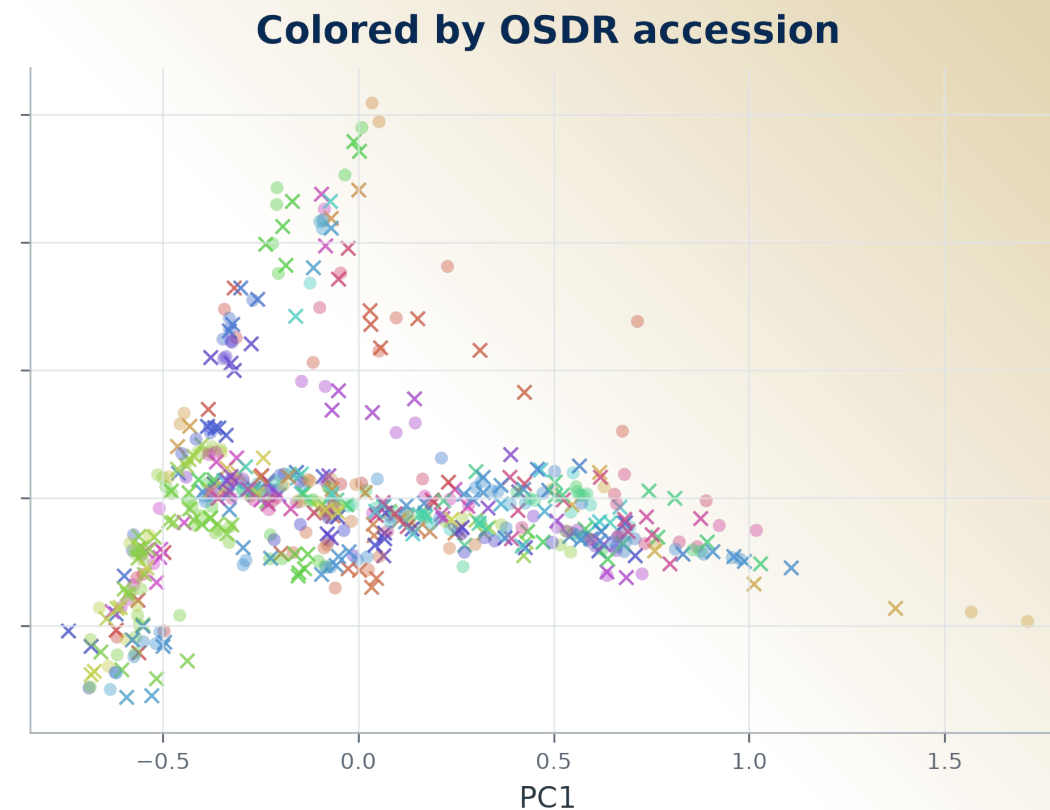
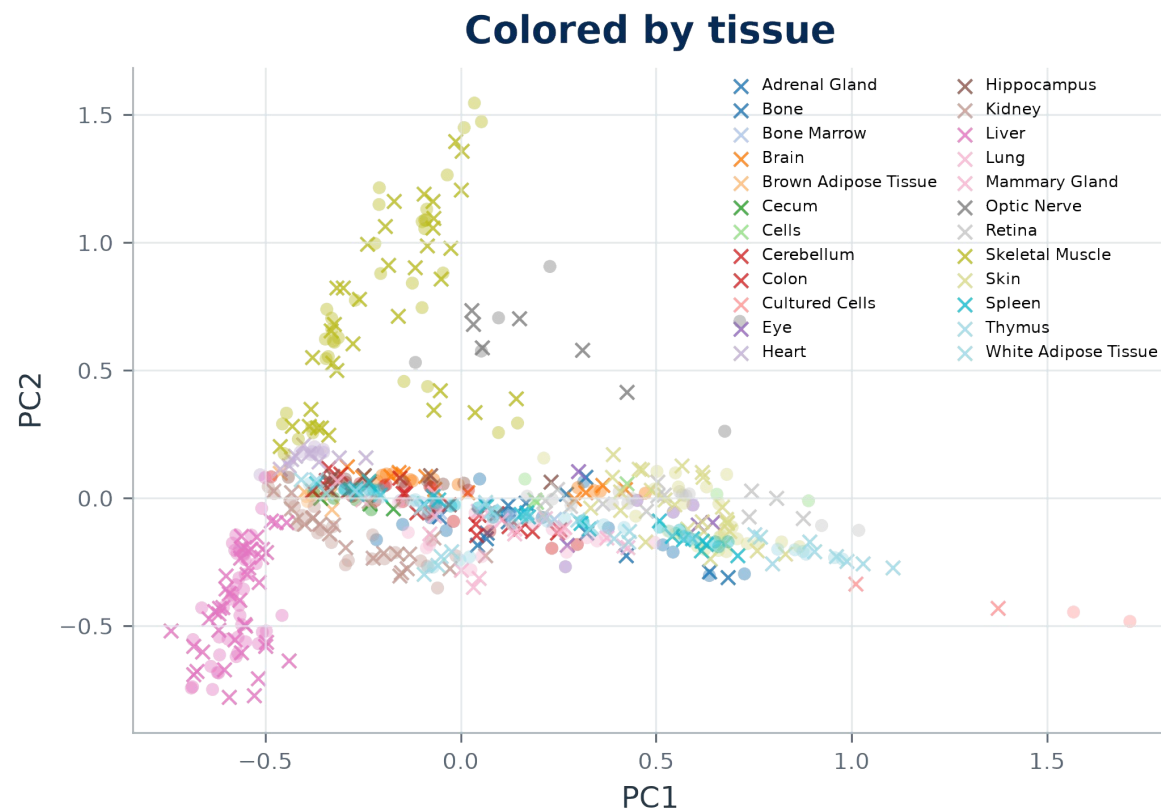
The same PCA axes follow 1,024 generated profiles through reverse diffusion.



DIFFUSION OUTPUT

Tissue and study structure dominate the PCA space

Left: tissue. Right: OSDR accession. Circles are observed profiles; crosses are matched DDIM profiles.

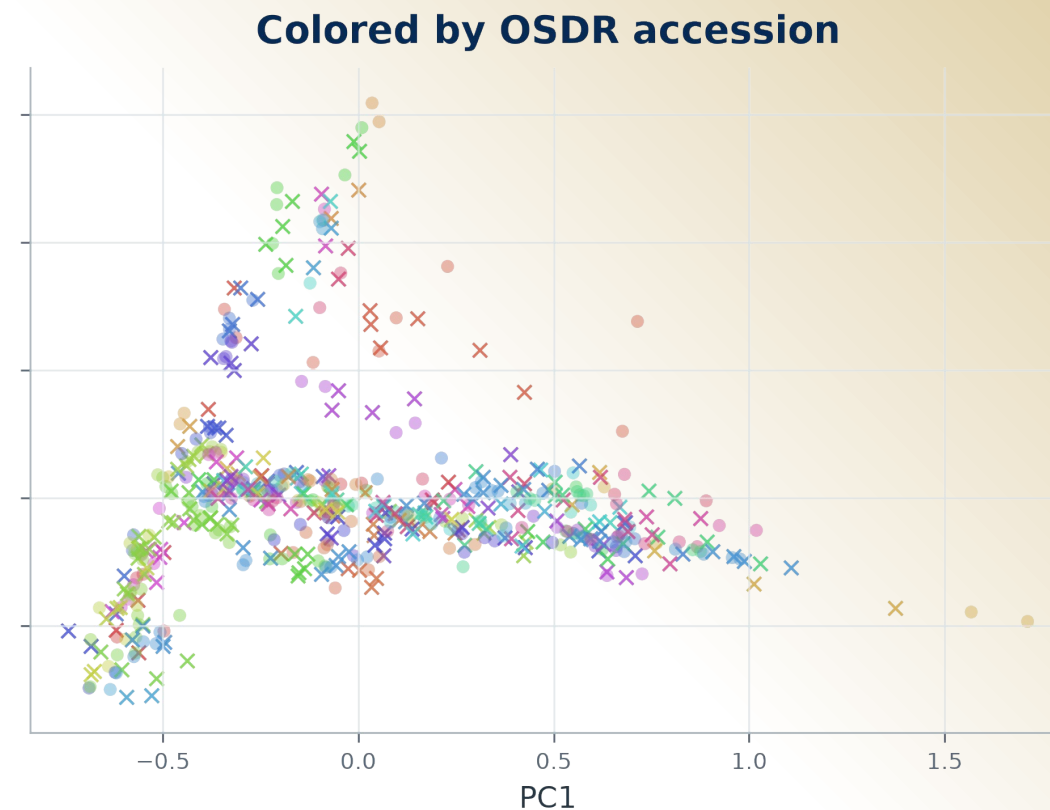
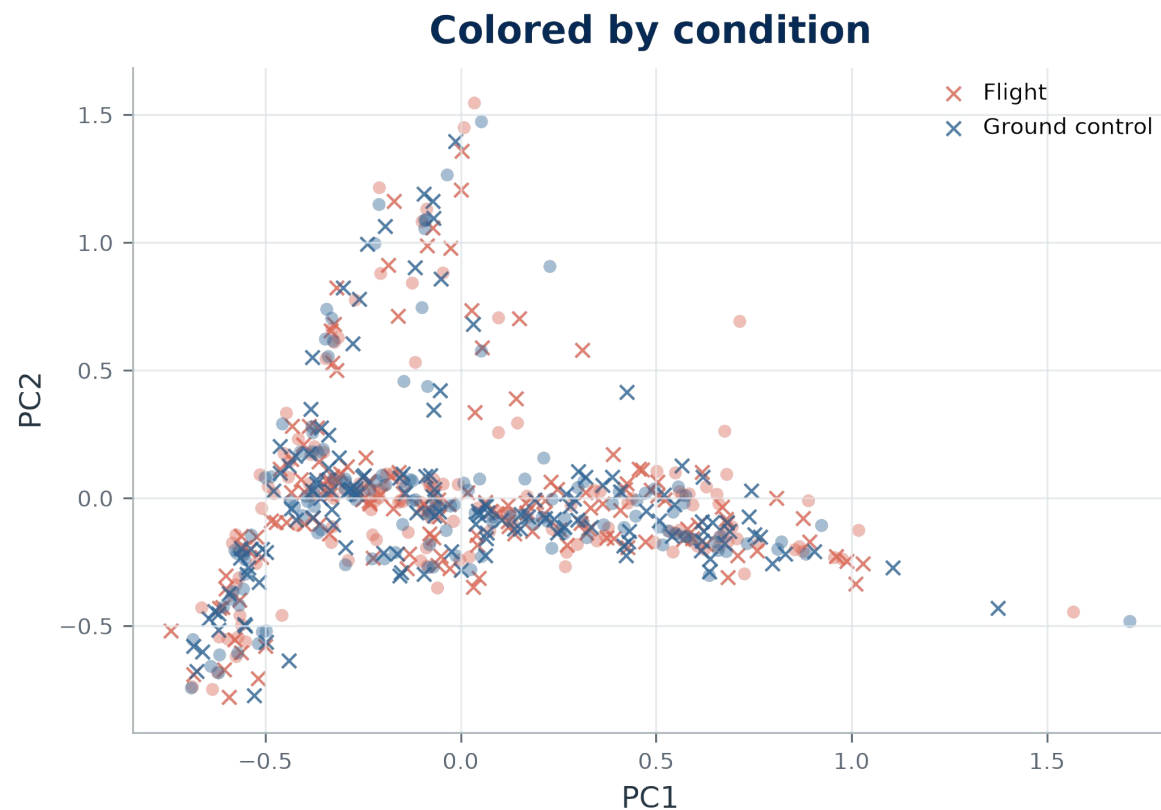


Tissue and accession clusters are visible across both observed and generated profiles.

DIFFUSION OUTPUT

Flight condition is subtler than study structure

Left: FLT or GC. Right: OSDR accession. Circles are observed profiles; crosses are matched DDIM profiles.



Flight and ground-control profiles overlap within the larger accession-defined structure.

ANALYSIS

Five arms separate gene ranking from classifier fitting

Guided arms use synthetic profiles to help choose genes; they do not treat generated profiles as new animals.

INPUTS	R real OSDR	S matched DDIM	GUIDED	R + S help rank genes	5%	S carries 5% of total fit weight
ANALYSIS ARM	GENE RANKING		CLASSIFIER FIT		WHAT IT TESTS	
Real only	R →	Rank from real	R →	Fit real	Baseline	
Generated only	S →	Rank from generated	S →	Fit generated	Synthetic-only stress test	
Real + generated	R + S →	Consensus rank	R + S →	Equal total weight	Direct augmentation	
Guided: real fit	R + S →	Consensus rank	R →	Fit real only	Feature guidance only	
Guided: 5% synthetic	R + S →	Consensus rank	R + S →	S = 5% weight	Guidance + light regularization	

All five arms → Held-out real profiles → BA | AUROC | AP → Choose an eligible arm within each tissue

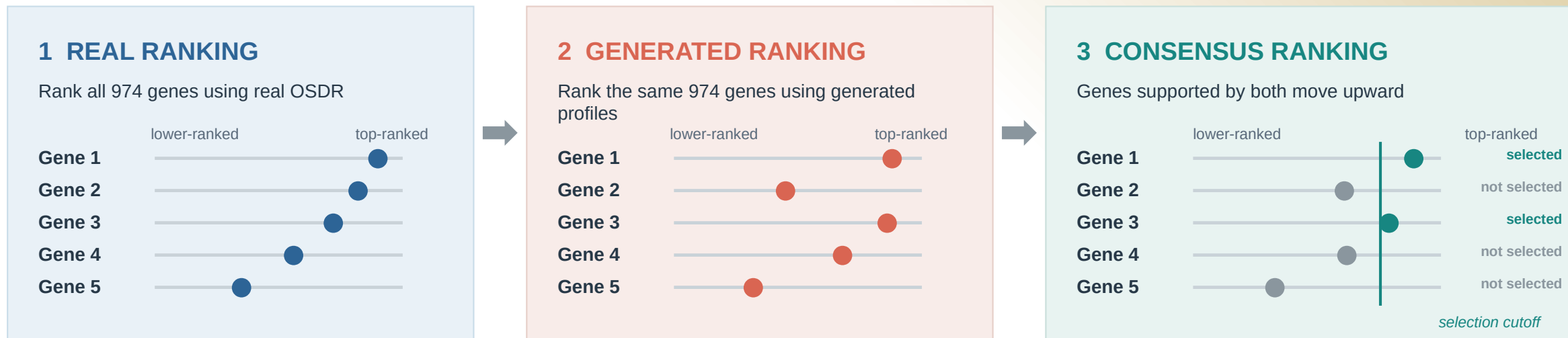
Association test FLT vs GC effects and BH FDR use real OSDR profiles only.

Animal n stays unchanged.

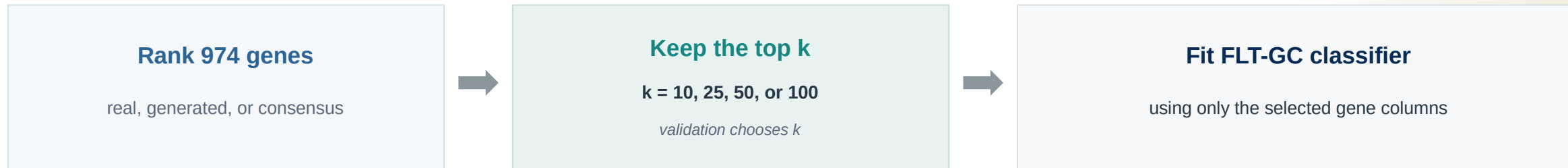
ANALYSIS

Consensus ranking chooses the classifier input genes

Real and generated rankings combine, then only the top k genes are used for FLT-GC classification.



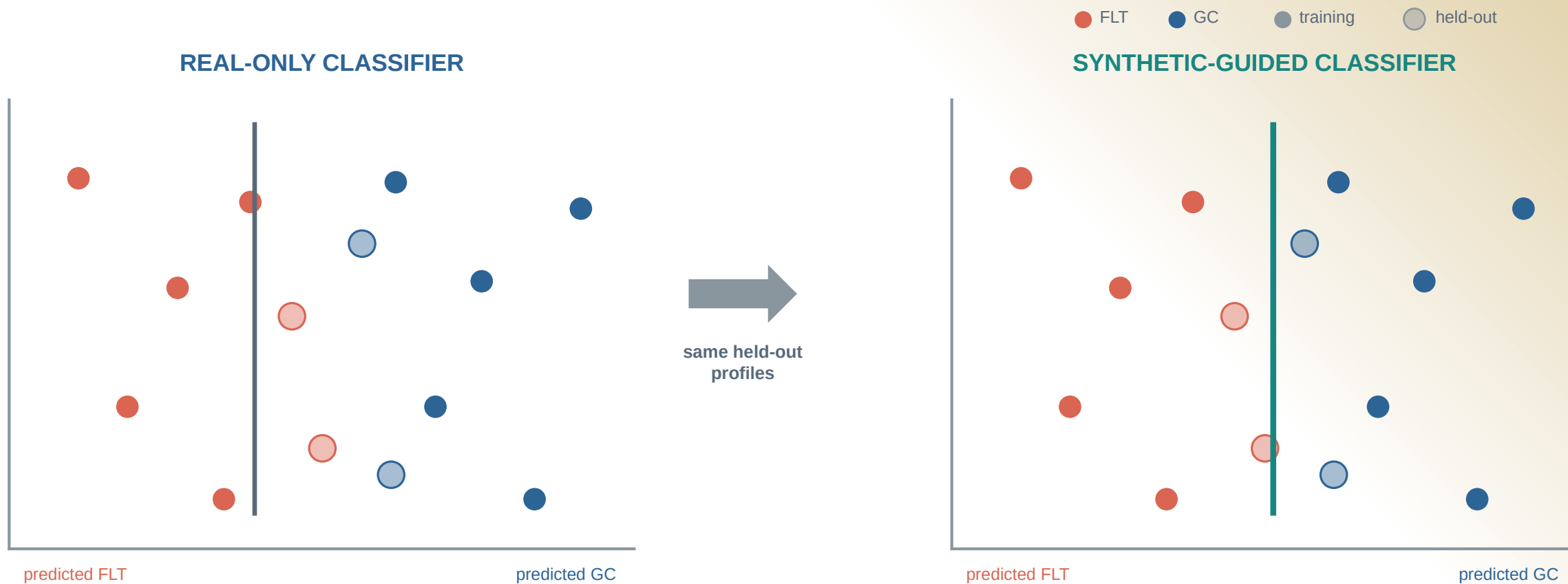
Top-ranked genes become classifier inputs



ANALYSIS

Synthetic guidance can shift the FLT/GC boundary

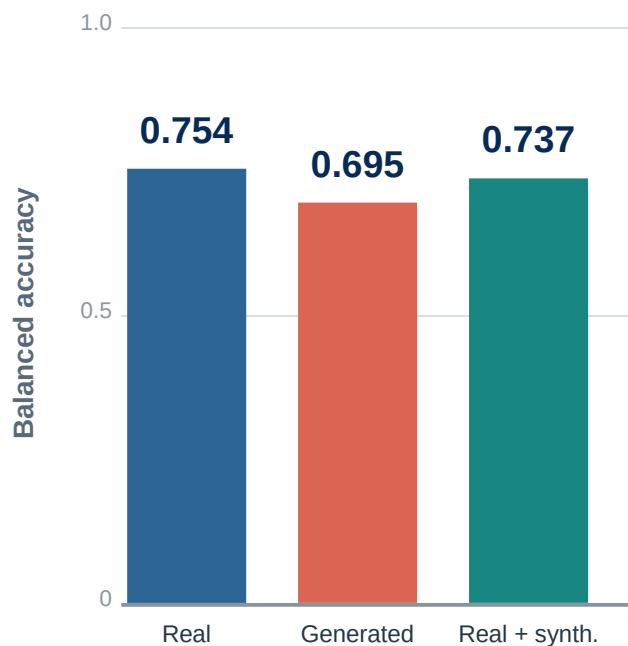
Opaque points train the classifier; transparent points are the same held-out real profiles in both panels.



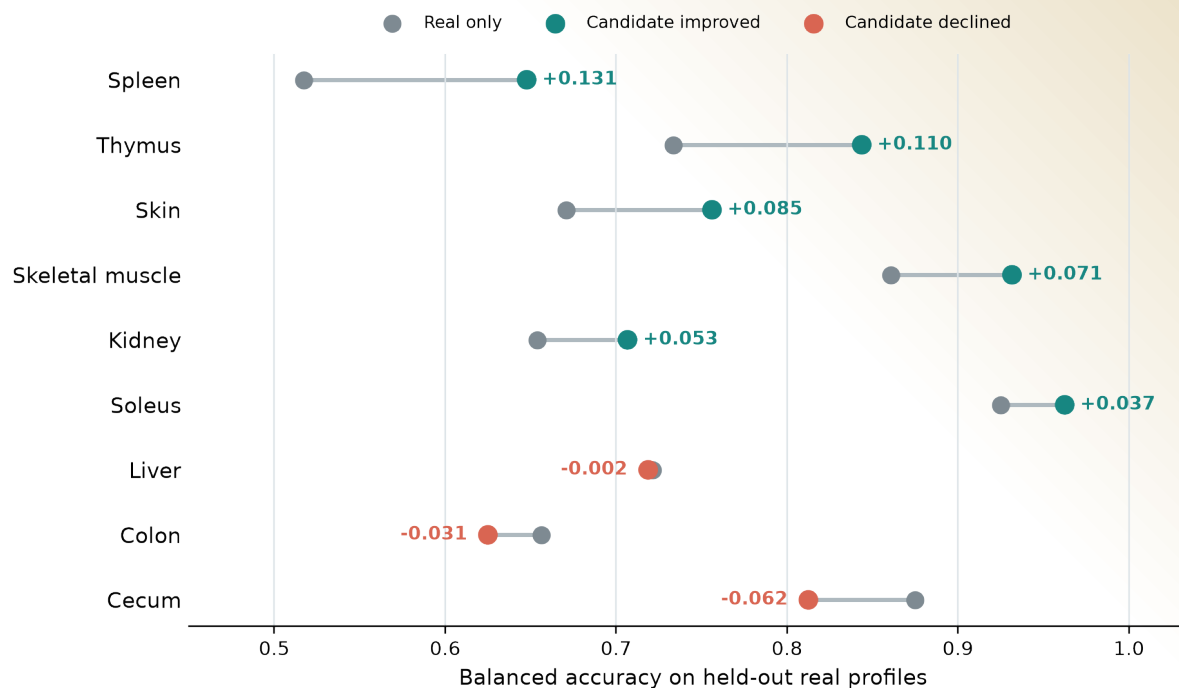
Synthetic use helped some tissues and hurt others

The pooled classifier declined, and even the best tissue-specific synthetic candidate was not always better than real-only training.

Pooled across tissues



Best synthetic-informed candidate by tissue



Only candidates that were non-worse on BA, AUROC, and AP were retained.

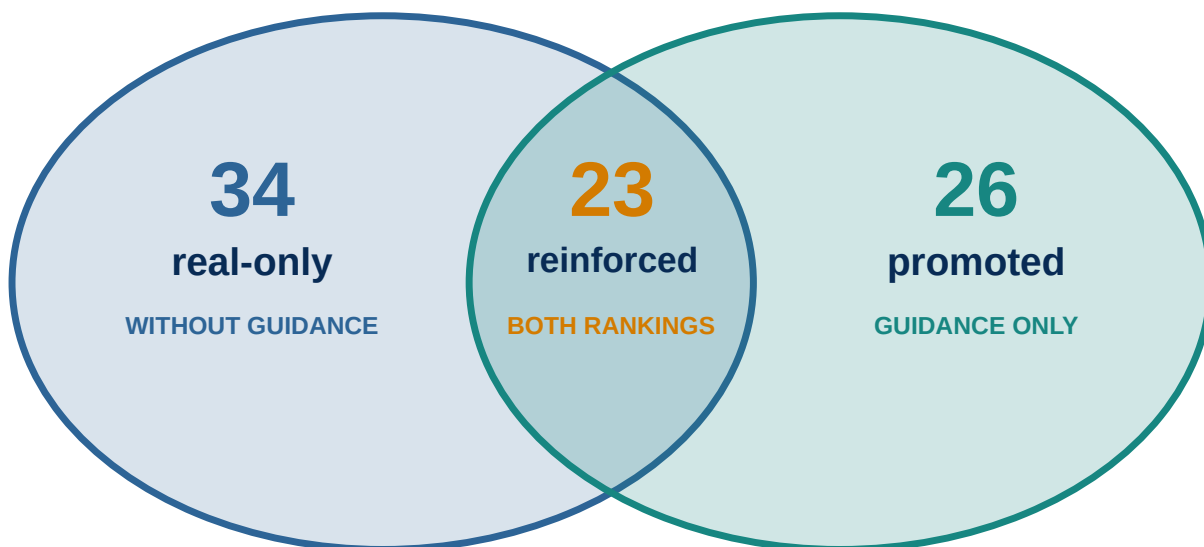
FEATURE ANALYSIS

Synthetic guidance changed ranking, not statistical evidence

Blue marks real-only selection; teal marks synthetic-guided selection. Every displayed association has real OSDR support.

Stable selection after real-data support

- Blue: stable in real-only ranking ● Teal: stable in synthetic-guided ranking



Real OSDR evidence gate

- Estimate FLT vs GC inside each accession
- Combine accession effects with a random-effects model
- Apply BH FDR within each tissue

23 reinforced + 26 promoted
49 synthetic-informed associations

The 34 real-only associations remain real-data findings; synthetic guidance did not reinforce them.

LITERATURE INTERPRETATION

Selection and literature are separate dimensions

All 49 synthetic-informed associations received both a selection status and a literature interpretation.

01 Fix the candidate

Gene, tissue, and FLT direction

02 Search

Spaceflight and mechanism literature

03 Compare

Observed result with published evidence

04 Record

Category, rationale, and source IDs

Selection status

26**Promoted**

Stable only with synthetic guidance

23**Reinforced**

Stable with and without guidance

Literature interpretation

Class	Count	Meaning
■ Aligning	22	Direct or same-tissue process agreement
■ Complementary	19	Related process or mechanism
■ Ambiguous	4	Mixed or context-dependent evidence
■ Unmatched	4	No sufficiently specific match found

A reinforced gene can be complementary, ambiguous, or unmatched; an aligning gene can be promoted.

ANALYSIS COVERAGE

The screen covered all 27 completed tissue analyses

The biological discussion narrows only after reporting synthetic-informed, real-only, and null outcomes.

10 Synthetic-informed

Promoted or reinforced BH-FDR gene

ANALYSIS UNITS

- Adrenal Gland
- Eye
- Kidney
- Skeletal muscle (pooled)
- Skin
- Spleen
- Thymus
- Gastrocnemius
- Soleus
- Tibialis Anterior

5 Real BH-FDR only

Real association, no synthetic-informed selection

ANALYSIS UNITS

- Heart
- Liver
- Retina
- EDL
- Quadriceps

12 No BH-FDR gene

No BH-FDR gene in the 974-gene panel

ANALYSIS UNITS

- Bone
- Bone Marrow
- Brain
- Brown Adipose Tissue
- Cecum
- Cerebellum
- Colon
- Hippocampus
- Lung
- Mammary Gland
- Optic Nerve
- White Adipose Tissue

GENE INVENTORY

Ten tissue analyses contained synthetic-informed genes

All 49 associations passed BH FDR in real OSDR data; synthetic profiles affected prioritization, not the statistical test.

Rows: FLT direction + status

aligning

complementary

ambiguous

unmatched

Gene color = literature | Non-empty rows shown

Thymus 16 genes	FLT higher Promoted Hsd17b11, Etv1	
	FLT higher Reinforced Snx7	
	FLT lower Promoted Nusap1, Stmn1, Birc5, Cdk1, Top2a, Ccnb2, Aurka, Ccne2, Kif20a, Pcna, Ccnf	
	FLT lower Reinforced Ube2c, Gmnn	
Spleen 4 genes	FLT higher Promoted Rai14, Myl9, Ptprk	
	FLT higher Reinforced Loxl1	
Kidney 2 genes	FLT higher Promoted Inpp4b	
	FLT higher Reinforced Slc37a4	
Gastrocnemius 2 genes	FLT higher Promoted Nfkbia	
	FLT lower Promoted Fhl2	
Eye 1 gene	FLT lower Reinforced Klhl21	
Skeletal muscle (pooled) 12 genes	FLT higher Reinforced Sox4, Cebpd, Sh3bp5, Prkcd, Arid5b, Sesn1, Tle1	
	FLT lower Promoted Klhl21, Mapkapk5, Reep5, Itgb5	
	FLT lower Reinforced Bphl	
Soleus 5 genes	FLT higher Reinforced Tpm1	
	FLT lower Reinforced Bdh1, Ech1, Bnip3, Decr1	
Tibialis anterior 4 genes	FLT higher Promoted Cebpd	
	FLT higher Reinforced Cdkn1a, St3gal5, Bnip3	
Adrenal gland 2 genes	FLT lower Promoted Psmb8	
	FLT lower Reinforced Tspan4	
Skin 1 gene	FLT higher Promoted Plscr1	

Separate rows report FLT direction and selection status; gene color independently reports literature interpretation for all 49 associations.

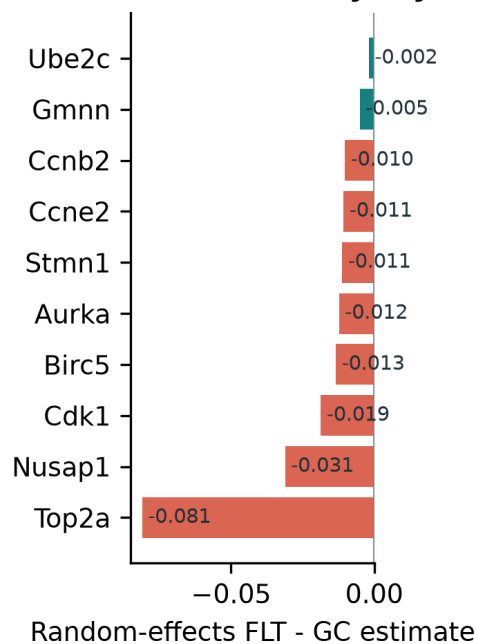
THYMUS

Thymus points to lower proliferative renewal

The cell-cycle program persisted without material conditioning; two FLT-higher genes extend the hypothesis.

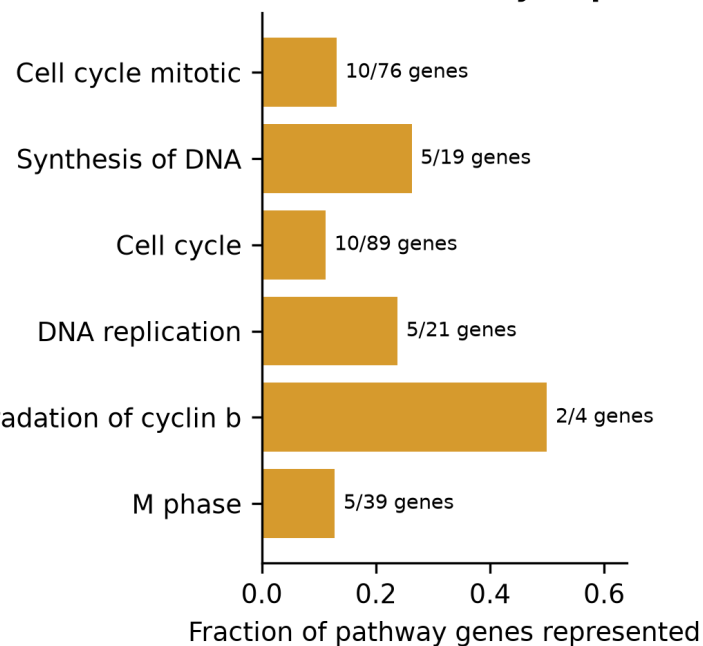
Synthetic-informed thymus genes converge on a flight-lower mitotic program

A Cross-study thymus effects



APC/C CDC20 mediated degradation of cyclin b

B Shared cell-cycle processes



16

synthetic-informed
BH-FDR associations

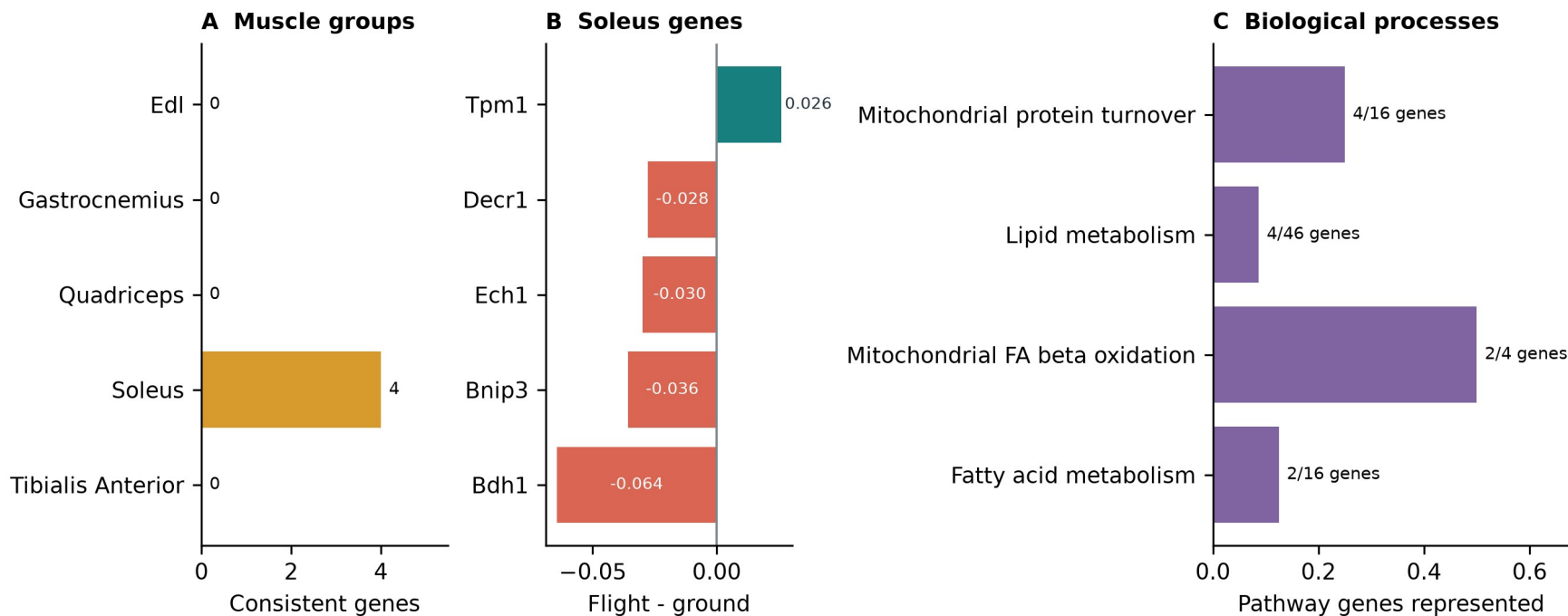
13 promoted | 3 reinforced

- Lower Nusap1, Stmn1, Birc5, Cdk1, Top2a and related genes
- Cell cycle, DNA replication, APC/C and G2/M processes
- Higher Hsd17b11 and Etv1 suggest lipid or T-cell-state shifts
- Bulk RNA-seq cannot separate cell loss from lower transcription

Soleus reinforces a mitochondrial and lipid program

The real-data program is coherent, but its synthetic reinforcement required explicit material conditioning.

Anatomical separation reveals a soleus-specific oxidative-metabolism hypothesis



0.925 to 0.963

balanced accuracy

5 reinforced | 0 promoted

- Lower Bdh1, Ech1, Bnip3 and Decr1
- Higher Tpm1
- Mitochondrial turnover and fatty-acid metabolism
- Compatible with unloading and contractile remodeling

ADDITIONAL FINDINGS I

Additional tissues produced distinct hypotheses

Promoted and reinforced selections are shown separately for each tissue.

Analysis unit	Selection	Genes and FLT direction	Interpretation
Pooled muscle	Promoted	Lower: Klhl21, Mapkapk5, Reep5, Itgb5	Heterogeneous stress, differentiation, interferon and sialic-acid response; interpret with the anatomical muscle groups.
	Reinforced	Higher: Sox4, Cebpd, Sh3bp5, Prkcd, Arid5b, Sesn1, Tle1; lower: Bphl	
Kidney	Promoted	Inpp4b higher	Renal phosphoinositide signaling and glucose-handling hypothesis; the pair had no shared Reactome enrichment.
	Reinforced	Slc37a4 higher	
Spleen	Promoted	Rai14, Myl9, Ptprk higher	Adhesion, actomyosin and extracellular-matrix or immune-organization hypothesis; no coherent pathway enrichment.
	Reinforced	Loxl1 higher	
Skin	Promoted	Plscr1 higher	Interferon-linked skin candidate within broader cell-cycle and DNA-repair responses; a single-gene result.
	Reinforced	None	

ADDITIONAL FINDINGS II

Eye, adrenal and muscle-group results remain tissue-specific

Promoted and reinforced selections are separated within every tissue.

Analysis unit	Selection	Genes and FLT direction	Interpretation
Eye	Promoted	None	Process-level alignment with lower proliferation and cytokinesis in flight eye tissue; not an exact prior gene replication.
	Reinforced	Klhl21 lower	
Adrenal gland	Promoted	Psmb8 lower	Literature-unmatched immune, proteostasis or tissue-composition candidates; no adrenal mechanism is established.
	Reinforced	Tspan4 lower	
Gastrocnemius	Promoted	Nfkb1a higher; Fhl2 lower	NF-kappaB stress alignment plus an autophagy or myogenesis candidate; the two genes do not form a coherent pathway.
	Reinforced	None	
Tibialis anterior	Promoted	Cebpd higher	Stress, cell-cycle, ganglioside-signaling and mitophagy candidates with mixed or complementary prior evidence.
	Reinforced	Cdkn1a, St3gal5, Bnip3 higher	

TAKEAWAYS

Synthetic data worked best as a tissue-specific prior

The generator helped prioritize observed signal. Biological claims still come from OSDR profiles.

01 Generate

Conditional DDIM produced high-fidelity profiles with near-chance real-versus-synthetic separation.

02 Use

Tissue-specific consensus ranking and light synthetic regularization improved held-out prediction in selected tissues.

03 Interpret

Promoted and reinforced genes supported tissue-specific hypotheses. Literature review separated prior alignment from complementary findings.

Thank you

Questions?

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Mentor

James Casaletto

Program

NASA Space Life Sciences Training Program

Data

NASA OSDR | ARCHS4 | Reactome

Compute

NASA Ames Research Center

Code and
manuscript

github.com/jasont314/nasa-mouse

All biological associations were tested in observed OSDR profiles.